MIL-D-40605C
5 December 1978
SUPERSEDING
MIL-D-40605B
27 December 1967

MILITARY SPECIFICATION

DISPLAY CASES, MECHANICALLY REFRIGERATED, SELF-SERVICE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers refrigerated, self-service, food display cases.

* 1.2 Classification. Display cases covered by this specification shall be of the following types, styles, classes, superstructures (non-refrigerated), suffixes, and sizes, as specified (see 6.2). The combination of available styles, classes, superstructures, and suffixes are covered herein:

Types

Type I - Produce
Type II - Dairy Products
Type III- Meat
Type IV - Frozen Food
Type V - Ice Cream

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4110

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Styles

Style A - Single deck - Wall installation
Class 1 - Without additional refrigerator shelves.
Class 2 - One additional refrigerator shelf.
Class 3 - Two additional refrigerator shelves.
Class 4 - Three additional refrigerator shelves (air curtain).
Class 5 - Four additional refrigerator shelves.
Class 6 - Five additional refrigerator shelves (no air curtain).

Style B - Single deck, center aisle installation.

Style C - Narrow twin deck, center aisle installation.

Style D - Wide twin deck, center aisle installation.

Style E - Multi deck, dairy-deli, end display.
Class 4 - Three adjustable shelves.

Style F - Single deck, air curtain cooler.
Class 4 - Three additional refrigerator shelves.

Style G - Bottomless cooler with roll-in carts.

Superstructure (Sup) (Not refrigerated).

Sup M-0 - No superstructure required.
Sup M-1 - High canopy, fixed, slanted mirror (available for only Style A, Class 1 in all types, except II)
Sup M-2 - Low canopy, fixed, slanted mirror (available only for Style A, Class 1 in all types, except II)
Sup M-3 - With sliding, slanted mirror (available only for Style A, Class 1 in all types, except II)
Sup M-4 - With sliding, vertical, clear glass (available only for Style A, Class 1 in all types, except II)
Sup M-5 - High canopy, with shelves (available only for Style A, Class 1 in all types, except II)
Sup M-6 - Low canopy, with shelves (available only for Style A, Class 1 in all types, except II)
Sup M-7 - Truss type shelves (available for type IV and V in Style C and D only)

Suffix SF - Storage below deck - Front access door (s) (available for type I, Style A and type III, Style A only)

Suffix SR - Storage below deck - Rear access door (s) available for type I, Style A and tape III, Style A only)

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Suffix HRLD - Vertical insulated back with insulated hinged rear loading doors. (types I through III only)

Suffix SRLD - Vertical insulated back with noninsulated sliding doors for loading from refrigerated area (types I through III only).

Sizes

Size 2 - For 2 roll-in racks (type II, Style G only)
Size 3 - For 3 roll-in racks (type II, Style G only)
Size 5 - 5 foot long (types II through IV Styles B and E only)
Size 6 - 6 foot long (types II and III, Style A, class 6 and types IV and V, Styles C, D, and E)
Size 8 - 8 foot long (all types and all Styles)
Size 12 - 12 foot long (all types and all Styles)

Ambient temperature

Air-conditioned area - 78° F (25°C) and 50 percent relative humidity
Non air-conditioned area - 90°F (32°C)

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

BB-F-1421	-	Flurocarbon Refrigerants
TT-P-664	-	Primer Coating, Synthetic, Rust-Inhibiting, Laquer-Resisting
MMM-A-260	-	Adhesive, Water-Resistant, (for Sealing Waterproofed Paper)
PPP-B-621	-	Boxes, Wood, Nailed and Locked-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-1055	-	Barrier Material, Waterproofed, Flexible

MILITARY

MIL-C-104	-	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	-	Preservation, Packaging Methods of
MIL-L-10547	-	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible
MIL-R-43979	-	Rack, Milk Container, Mobile

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 117 - Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 471 - Commercial Refrigerators and Freezers

(Application for copies should be addressed to the Underwriters Laboratories Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Rd., Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, LI, NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050; or 2602 Tampa East Blvd., Tampa, FL 33619.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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echnical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The display cases covered by this specification shall be the current standard product of the manufacturer except for any design modifications specified herein.

3.2 Safety standard. The display cases, including display cases with open type compressors and display cases with remote condensing units, shall conform to UL 471.

3.2.1 Certification. Prior to approval of the first shipment, the contractor shall submit to the contracting officer or his authorized representative, the satisfactory evidence that the display cases he proposes to supply under this specification meet the applicable requirements of UL 471. Acceptable evidence of meeting the applicable requirements of this standard shall be the UL label, an approved UL listing, or a certified test report from a recognized independent testing laboratory, indicating that the display cases offered conform to the applicable requirements of UL 471.

3.3 Materials. The materials shall conform to the applicable specifications and requirements specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer for the purpose in commercial practice. Commercial tolerances shall apply to all metal thicknesses and diameters unless otherwise specified herein (see 6.4).

3.3.1 Insulation. Insulation shall be industrial type fiberglass, expanded polystyrene, or expanded polyurethane, shall be homogeneous throughout, and shall have a "K" factor rating determined by using a 75°F (24°C) mean temperature and expressed in British thermal units (BTu) per hour per square foot per degree temperature difference per inch of thickness (watt per metre Kelvin). Loose fill type insulation shall not be acceptable. Flame retardant grades shall be used where needed to meet requirements of UL 471.

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* 3.3.1.1 Rigid. Rigid insulation shall be pre-expanded polyurethane or pre-expanded polystyrene having a manufacturer's commercial rated "K" factor of not more than 0.15 (0.02 W/m·K) for polyurethane and 0.25 (0.04 W/m·K) for polystyrene and a minimum core density of 1.8 pounds per cubic foot (29 kg/m^3) (see 4.4.7).

* 3.3.1.2 Foamed-in-place. Foamed-in-place insulation shall be polyurethane foam having a manufacturer's commercial rated "K" factor of not more than 0.15 (0.02 W/m·K). The foam shall have a density of not less than 1.7 pounds per cubic foot (27 kg/m^3) when froth and 1.9 pounds per cubic foot (30 kg/m^3) when poured with density tested as specified in 4.4.6.

* 3.3.1.3 Semirigid. Semirigid insulation shall be "B" fiber or finer industrial type fiberglass with a "K" factor rating of not more than 0.25 (0.4 W/m·K) and with a commercial manufacturer's rated density of at least 1.0 (16 kg/m^3) but not more than 3.0 pounds per cubic foot (48 kg/m^3) and shall be held together with a thermosetting binder.

* 3.3.2 Refrigerant. Refrigerant used in the refrigerating system of the display cases shall conform to BB-F-1421.

3.4 Design and construction.

3.4.1 Design. Display cases shall be designed for merchandising the specific types of products as indicated in 1.2 and the types, styles, superstructures, classes, suffixes, sizes and temperature operations specified to be furnished. When a superstructure is to be furnished as part of display case, the display case shall be designed and fabricated to accept and support the superstructure and create an appearance of an integral unit. Shelves, racks, lighting, price tag moulding, heat strips, cart bumpers, splash guard (kick-plate), drain, and trim shall be furnished and installed as specified in accordance with the contractor's practice for a complete installation. The display cases for free standing placement shall have readily removable insulated ends and where lineups are specified, shall be designed and furnished with one set of common ends for each lineup, partitions, and fittings to allow for matching lineup installations of the same type cases. Front glass guards for front sill shall be provided on cases in accordance with manufacturer's standard practice. Unless otherwise specified (see 6.2), all glass shall be tempered glass or safety glass.

* 3.4.2 Construction. Display cases shall be constructed to provide positive refrigeration and protection of contained foods in both the display and storage areas in the specified temperature range. The exterior shall be constructed to facilitate cleaning, inspection, servicing, and maintenance. The panel joints and seams shall be sealed and made smooth. Exposed rivets and screw or bolt heads shall be of the low profile type such as brazier or modified brazier rivets or pan and oval heads, respectively. The entire

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terior shall be arranged and assembled to be readily cleanable throughout. Removal of pans shall give accessibility for servicing and permit cleaning of coils and fans. Night lids shall be provided for ice cream and frozen food cases, and shall be fabricated to fit the cases. All lighting shall be recessed or shielded so only reflected light can be seen from the front of the case, and separate switching shall be provided for light circuits. Price tag mouldings to take 1-1/4-inch (32 mm) tags shall be installed on each case in accordance with manufacturer's standard practice. A junction box or boxes shall be installed for the lights, fans, heating strips, and defroster circuits and shall be easily accessible for maintenance and servicing. Heat strips shall be installed as required to prevent formation of condensation on the exterior front and top exposed sills. Heat strips shall also be installed to reduce condensation to a minimum on the other exterior surfaces of the display case. Bolts, clips, pins, locking cams, fasteners, accessories, common ends, sealing kits, or other parts required shall be furnished. The display cases shall be completely ready for installation or for field assembly when cases are shipped other than completely assembled for operation. Units to be assembled on the installation site shall be marked and furnished with complete assembly drawings and installation instructions.

3.4.3 Matching lineups. Display cases for matching lineup shall be provided with one set of ends (1L and 1R) for each matching lineup furnished as specified (see 6.2). Plugs and partitions shall be furnished with the display case, canopies, and superstructures when adjoining units have different operating temperatures; when single deck units adjoin double-duty cases; when adjoining units operate on different defrost cycles; or when otherwise essential for a complete installation in satisfactory operating condition. The plugs and partitions shall be the insulated or noninsulated type as required with plexiglass fillers or sections in exposed areas, and shall be furnished with the necessary fasteners and trim. Materials and finish of the plugs and partitions shall match adjoining sections in the display case and superstructures. Installations of the plugs and partitions shall be made by the contractor in the shop; and shall be shipped separately only when essential to remove items for crating and shipping or when the items cannot be installed until the cases are installed and fitted together. When plugs or partitions and trim are furnished as separate items for installation at the site, such items shall be marked and complete installation instructions and diagrams shall be inclosed with the items.

3.4.4 Physical characteristics and performance requirements.

* 3.4.4.1 Dimensional requirements. The dimensions shall be as specified in table I and II. The size 5 display case shall have a nominal length of 5 feet (1.5 m), the size 6 display case shall have a nominal length of 6 feet (1.8 m), the size 8 display case shall have a nominal length of 8 feet (2.4 mm), and the size 12 display case shall have a nominal length of 12 feet (3.6 m). The size 6 display case shall be used to close off ends for styles C and D.

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TABLE I. Display case dimensions, outside

Style	Class	Depth (inches) (m)	Height: Front (including glass) (inches) (m)	Height: Back (inches) (m)
A	1	39 to 48 (1 to 1.2)	28 to 42 (.7 to 1)	38 to 44 (.9 to 1.1)
A	2	39 to 48 (1 to 1.2)	28 to 42 (.7 to 1)	50 to 60 (1.2 to 1.5)
A	3	39 to 48 (1 to 1.2)	28 to 42 (.7 to 1)	56 to 75 (1.4 to 1.9)
A	4	39 to 48 (1 to 1.2)	15 to 35 (.3 to .8)	65 to 85 (1.6 to 2.1)
A	5	39 to 48 (1 to 1.2)	15 to 35 (.3 to .8)	65 to 85 (1.6 to 2.1)
A	6	25-1/2 to 34 (.6 to .9)	11 to 25-3/4 (.2 to .6)	79 to 85 (2 to 2.1)
B	-	39 to 47 (1 to 1.2)	28 to 42 (.7 to 1)	28 to 42 (.7 to 1)
C	-	56 to 72 (1.4 to 1.8)	28 to 42 (.7 to 1)	28 to 42 (.7 to 1)
D	-	73 to 87 (1.8 to 2.2)	28 to 42 (.7 to 1)	28 to 42 (.7 to 1)
E	4	40 to 44 (1 to 1.1)	25 to 29 (.6 to .7)	60 to 69 (1.5 to 1.7)
F	4	31 to 45 (.7 to 1.1)	15 to 33 (.3 to .8)	70 to 82 (2 to 2)
G	-	44 to 48 (1.1 to 1.2)	11 to 15 (.2 to .3)	80 to 85 (2 to 2.1)

TABLE II. Superstructures (nonrefrigerated), outside

Superstructures	Top height above floor (inches) (m)	Depth at top front to back (inches) (m)
M-1	67 to 75 (1.7 to 1.9)	20 to 30 (.5 to .7)
M-2	55 to 67 (1.3 to 1.7)	15 to 21 (.3 to .5)
M-3	67 to 75 (1.7 to 1.9)	20 to 30 (.5 to .7)
M-4	69 to 77 (1.7 to 1.9)	15 to 26 (.3 to .6)
M-5	62 to 75 (1.5 to 1.9)	20 to 30 (.5 to .7)
M-6	55 to 60 (1.3 to 1.5)	15 to 21 (.3 to .5)
M-7	56 to 64 (1.4 to 1.6)	20 to 30 (.5 to .7) (see 3.4.6.3)

* 3.4.4.2 Internal temperature requirements. The display cases shall be designed, constructed, and equipped with refrigerating systems so that the internal display case temperatures shown in table III shall be maintained in one of the display case ambient operating temperatures of 78°F (25°C) and 50 percent relative humidty or 90°F (32°C) as specified (see 6.2). (Night covers on type IV and type V cases may be used to meet the requirement of operation in 90°F (32°C) ambient.) For remote units the condensing unit shall be operated in an ambient of 100°F (38°C). When tested as specified in 4.4.1.2, display cases with either 60 or 50-hertz motors shall maintain internal temperatures within the limits specified in table III with the

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running time of the condensing unit not to exceed 75 percent for 78°F (25°C) display case ambient temperature; and 85 percent for 90°F (32°C) display case ambient temperature. During conduct of the test required by 4.4.1.2, the exterior surface of the display case shall be not more than 10°F (6°C) below ambient temperature (see 4.4.2). Satisfactory operation in more than one of the ambient test conditions is not a requirement of this specification. Remote condensing units for display cases shall operate in an ambient temperature of 110°F (43°C) without cutting out on motor overload, and self-contained display cases shall operate in 100°F (38°C) ambient temperature without cutting out on motor overload (see 4.4.1.2b).

TABLE III. Display and storage area temperature

Type	Minimum	Maximum
I	32°F (0°C)	47°F (8°C)
II	31°F (-0.5°C)	42°F (5°C)
III	28°F (-2°C)	38°F (3°C)
IV	None	0°F (-18°C)
V	None	-8°F (-22°C

3.4.4.3 Operation. Display cases shall be operational when tested as specified in 4.4.4. The refrigerant systems shall not leak, electrical equipment shall be connected without shorts or grounds, all equipment shall be adjusted as required; and all controls shall function.

3.4.5 Display cases styles.

3.4.5.1 Double-duty cases. Display cases designated by SF or SR for double-duty shall have provision for refrigerated storage in the bottom of the case. Doors shall be provided for access to the storage space. Access doors shall be in the front or in the rear as specified (see 6.2) and may be hinged or sliding.except that aisle doors on front of cases shall be sliding type. Doors shall be insulated. Sliding type doors shall operate on stainless steel, rigid extruded plastic, or extruded aluminum tracks. Doors shall slide easily and shall be readily removable. Hinged doors shall have adjustable, chromium plated, brass hardware. The bottom storage area shall removable, sectional, full length center shelf and floor racks. Shelves shall be supported on guides to permit shelves to be inserted or removed easily. Guides shall hold the shelves in alignment so that shelves are not displaced or allowed to fall out of position. Floor racks shall be formed and supported to allow for air circulation beneath the stored products. Door liner and outer shell shall be finished and constructed to match the inner liner and outer shell of the display case.

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3.4.5.2 Style A, class 1. Display cases shall be single deep well deck for maximum depth of display and shall have adjustable racks for at least three positions to permit shallow display of various depths. Racks for types I and III shall be adjustable for horizontal or slanted display. Wall installations shall have low fronts for maximum display.

3.4.5.2.1 Style A, class 2. The class 2 display cases shall be designed with a lower deck and one additional shelf within the refrigerated area and for wall installation. The top of the case shall be insulated and designed as a shelf for display of nonrefrigerated items. The ends of the case shall be equipped with nominal 1/4-inch (6 m) tempered or safety glass screen wings where required to maintain refrigerated temperatures in the case. Shelves within the refrigerated area shall be adjustable up and down or shall be designed for setting horizontally in a fixed position or for angling up or down from front to back. The class 2 display case may be specified for all types, except type IV and V.

3.4.5.2.2 Style A, class 3, The class 3 display cases shall be the same as the class 2 display case except that the class 3 display case shall have two additional shelves within the refrigerated area. The class 3 display case may be specified for all types.

3.4.5.2.3 Style A, class 4. The class 4 display cases shall be designed for 80°F (27°C) ambient (air conditioned) spaces, with a lower deck, and three additional shelves for merchandising products behind a refrigerated air curtain. The class 4 display cases may be specified for all types. The class 4 shall be for wall installation.

3.4.5.2.4 Style A, class 5. The class 5 display case shall be the same as the class 4 except that the class 5 display case shall have four additional shelves within the refrigerated area.

* 3.4.5.2.5 Style A, class 6. The class 6 display case shall be the same additional shelves within the refrigerated area. The class 6 display case may be specified for all types, except type IV and type V.

3.4.5.3 Style B. The style B display cases shall be a single deep well deck and shall have adjustable racks for at least three positions to permit shallow displays of various depths. The style B display cases shall be for center aisle installation with a uniform top height front and back. The style B display cases may be specified for all types, except type I and II.

3.4.5.4 Style C. The style C display cases shall be a narrow twin single deep well deck having an overall depth not exceeding 72 inches (1.8 m), and shall have adjustable racks for at least three positions to permit shallow displays at various depths. The display area may be one double width compartment in lieu of two single width compartments. The style C display cases shall be for center aisle installation with a uniform height front and back. The style C display cases may be specified only for types IV and V.

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3.4.5.5 Style D. The style D display case shall be a wide twin single deep well deck the same as style C, except the overall depth shall be between 73 and 87 inches (1.8 to 2.2 m) and the display area shall be two separate compartments. The style D cases may be specified only for types IV and V.

3.4.5.6 Style E. The style E display case shall be a utility case for end display of meats or dairy products and shall have up to five adjustable shelves.

3.4.5.7 Style F. The style F display case shall be a single deck air curtain case designed to be installed into an existing cooler. The style F display case shall have three rows of adjustable shelves and rear loading doors or curtains. The curtains shall be of the type and style used in the manufacturer's commercial products. Doors shall be hinged or sliding type. The hinged doors shall be insulated and the sliding doors noninsulated. An end kit shall be provided to effect a smooth blend of air curtain with existing cooler. The type and size of ends shall be as specified (see 6.2). When two or more style F display cases are to be joined a joint kit shall be specified (see 6.2). The style F display case shall have lights standard with the manufacturer. When specified (see 6.2), the bottom deck shall be equipped with two position, adjustable, wire shelves.

3.4.5.8 Style G. The style G display case shall be a bottomless air curtain case designed for rear loading of roll-in racks. The style G display case shall be designed as a type II case (see 1.2) with canvas sliding curtain enclosures. When specified (see 6.2), a removable shelf kit to be used in lieu of one or two of the roll-in racks shall be furnished. The size 2, style G display case shall be furnished with 2 roll-in racks and the size 3, style G display case shall be furnished with 3 roll-in racks. The roll-in racks shall conform to MIL-R-43979.

3.4.6 Superstructures. Superstructure canopies shall be furnished with the display cases complete with light fixtures, trim and accessories, and shall be finished in colors for parts normally colored matching the display case on which it is to be mounted. Finish shall be baked enamel. Canopy superstructures shall be all steel or aluminum construction including back panel, shelves, reinforcing brackets, and supports; and shall be furnished as an assembled unit or as a prefabricated unit ready for assembly. Fluorescent light fixtures, with reflectors and bulbs, shall be installed the full length of canopy top for M-5, M-6, and M-7 superstructures. The top shelf may be considered the canopy top. A separate switch for control of canopy lights shall be provided. High and low shelf canopies supplied with display

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cases having a rear sill with less than 8-1/2 inches (216 mm) front to back of level display surface, shall be furnished with an extra 12-inch (304 mm) wide shelf designed to be installed approximately 3 inches (76 mm) above the sill. Canopies and shelves shall be supported, reinforced, and constructed to support a uniform load of not less than 50 pounds per square foot (244 $kg \cdot m^2$) of shelf without sagging or deforming below 1/4-inch (6 mm) (see 4.4.3), except that the loading does not apply to sliding glass or mirrored superstructure. Shelving for style M-7 shall be full length. End stops for 50 percent of shelf ends shall be furnished to close end of shelves to provide for shelf displays that are not continuous. Wire shelves shall be furnished for type I and flat metal shelves shall be furnished for other type cases as standard with the manufacturer.

* 3.4.6.1 Superstructures M-1, M-2, M-3, and M-4. Superstructures M-1, M-2, M-3, and M-4 shall extend the full length of the display case. Slanted mirrors shall be mounted at an angle to give full reflection of the display but shall not reflect any section of the floor from a straight ahead viewpoint. Slanted mirrors of high sliding type shall have an observation port 1-inch (25 mm) high across the width of each mirror. Sliding mirrors and clear glass shall be framed with stainless steel, chromium plated steel, or polished aluminum alloy, and shall be mounted in the display case on tracks and bearings that permit easy operation. Frames shall have chromium plated metal, aluminum, or stainless steel handles. Sliding mirrors and clear glass doors shall be easily removable for cleaning and replacement. Tracks for doors shall have drain holes. Impact protection at top points of opening and closing shall be provided. Mirrors and windows shall be plate glass. Superstructures M-1 and M-2 are available only for style A, class 1 in all types, except type II. Superstructures M-3 and M-4 are available only for style A, class 1 in all types of display cases, except type II.

3.4.6.2 Superstructure M-5 and M-6. Superstructure M-5 and M-6 shall extend the length of the display case and shall be the dimensions specified. Intermediate shelves shall be 12 to 14 inches (305 to 355 mm) deep. A bottom shelf shall be furnished as a part of the shelf canopy and shall be as specified in 3.4.6. All intermediate and bottom shelves shall be furnished in multiples of 4 feet (1.2 m) nominal length; shall be adjustable in height, 1-1/2 to 2 inches (38 to 51 mm) on centers; and shall be held solidly in place where installed. Shelves and support components including back plates shall be fabricated of the materials and finish specified in 3.4.6. Price tag mouldings to take 1-1/4-inch (32 mm) tags shall be installed for each shelf. Superstructures M-5 and M-6 are available only for style A, class 1 in all types of display cases, except type II.

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3.4.6.3 Superstructure M-7. Superstructure M-7 for style C and D for types IV and V display cases shall consist of center supported truss type top and bottom open shelves extending full length of the display case with pilaster supports 4 feet (1.2 m) nominal, on centers, or shall be of rigid construction designed to be supported by pilaster supports at each end. Shelf supports shall be braced and held rigidly in vertical position to insure stable shelving. The top shelf shall be approximately 22 to 24 inches (558 to 609 mm) above top sill of display case, and bottom shelf shall be 10 to 14 inches (254 to 355 mm) below top shelf. Bottom shelf shall be 12 to 14 inches (.3 to .35 m) deep overall for style C cases and 20 to 22 inches (508 to 558 mm) for style D cases. Top shelf shall be 20 to 24 (508 to 609 mm) inches deep overall for style C cases and 26 to 30 inches (660 to 762 mm) for style D cases. Price tag mouldings to take 1-1/4 inch (32 mm) tags shall be installed full length on both sides of top and bottom shelves.

3.4.7 Framing. Framing members shall form a rigid structure capable of providing support during handling and shipping, and for the loaded shelves, racks, liner, shell, trim, and superstructure. Frames shall be reinforced and predrilled for bolting or locking cases together as required for lineup installations and for twin aisle installations. Reinforcement and backing of the outer shell and inner liner shall be provided as required for support and fastening of hardware and brackets. Reinforcement and backing of the shell and liner shall be provided to support drain and eliminate flexing of outer shell, or inner liner during make-up of drain connections. Metal framing members shall be phosphatized steel finished with a rust and corrosion inhibitor, galvanized steel, aluminum, or corrosion-resisting steel. Reinforcing and framing members not totally inclosed, or within walls, shall be placed in a manner to be easily cleanable. The ends of all hollow sections of reinforcing and framing members shall be sealed. All vertical channel sections shall be either completely closed or open to the floor.

3.4.7.1 Support base. All display cases shall have rigid support bases fabricated from steel and recessed at the front of the case and shall be provided with recessed splash-guards (kick-plate). Metal shall be the same type as specified for framing members in 3.4.7.

3.4.8 Outer shell and inner liner. Access openings in the outer shell and inner liner shall be provided as required for assembly and installation. Clearance space and access openings shall be provided with fitted covers and shall be thoroughly sealed after fabrication to prevent moisture or water-vapor from entering the insulated space. Interior walls may have openings to permit vapor transmission from the insulation, provided such openings are located and protected to prevent the entry of free water from any source.

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All space between the outer shell and inner liner shall be filled with insulation of the types specified in 3.3.1.1, 3.3.1.2, or 3.3.1.3. Rigid or foamed-in-place insulation shall be used in the bottom space to support and to prevent any displacement of the liner, except that when low conductive supports are installed, semirigid insulation may be used.

* 3.4.8.1 Outer shell. The front, top, and ends of the outer shell shall be enameling iron, aluminum, or carbon steel. The bottom shall be stainless steel, aluminum, or galvanized steel. The rear shall be enameling iron, aluminum, carbon steel, or galvanized steel.

3.4.8.2 Inner liner. The inner liner or the insulated panels shall be aluminum alloy, fiberglass, or galvanized steel. When galvanized steel is used, the inner side shall not be exposed and shall be protected and concealed by the inside air baffles. Air baffles shall be plastic, aluminum, or steel finished with baked enamel, or shall be aluminum alloy except that inside front baffles (excluding style C and D) and lower deck baffles may be minimum spangled galvanized steel. Inside ends shall be of the materials specified for air baffles or as specified in 3.5.2. Seams shall be smooth so that interior of cabinet can be kept clean and sanitary. Tapping plates and structural reinforcement, as required, shall be secured to the insulation side of the inner liner to provide support for brackets, shelves, baffles, or other items to be fastened to, supported by, or hung on the exposed side of the inner liner. Duct openings in the display area shall be protected from accidental entrance of displayed items or foreign matter.

* 3.4.9 Drain. A drain pipe no smaller than 1-inch iron pipe size (I.P.S.) shall extend through the bottom of the display case. The drain hole and drain pipe shall be accessible for normal installation and cleaning without removal of any major components. The display case drain pan beneath the coils and the case bottom shall provide drainage so that free water does not stand on the drainage surfaces when cases are level. Drains shall be completely sealed to the case to prevent moisture from entering insulated space. All condensation shall be drained without dripping into storage areas. A length of 1-inch pipe, with formed trap located at least 18-inches (457 mm) from the drain nipple, shall be furnished with each type case. When a drain trap is built into the cases, the external trap is not required. The drain trap shall be brass or aluminum. Pipe and fittings shall be brass, aluminum, galvanized iron, or nylon.

3.4.10 Cart bumper. Display cases shall be provided with a cart bumper of stainless steel, aluminum, or high-impact-resistant styrene plastic. Cart bumpers may be used as a raceway for electrical conductors.

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* 3.4.11 Glass front. The front glass, when provided (see 3.5.1) for cases shall rise 2-1/2 inches (63 mm) to 10 inches (254 mm) above the top of the front sill. The glass used shall be 1/4-inch (6 mm) thick safety or tempered glass or shall be 2- or 3-glasspane sealed units, as required, to eliminate fogging. Glass fronts shall be capped with a channel section and supported vertically. All exposed channel section, supports, and trim for glass fronts shall be chromium plated brass, extruded and polished aluminum, or stainless steel. Glass fronts shall not project above case ends at their junction.

3.4.12 Hose and spray nozzle. Type I produce cases shall be equipped in the front part of the case with a water hose and a chromium plated brass nozzle to provide a fine mist spray. Hose shall be of sufficient length to service entire display area and shall be automatically retractable to a retainer in concealed position. Copper water supply service connection shall be extended from hose to exterior of case and terminated with a brass shut-off gate valve located in a recess within the outline of the display case.

3.4.13 Bottom storage (double duty cases). The bottom storage area for type III cases shall be constructed with a center shelf and with floor racks. Shelf and racks for storage area shall be steel, hot tin dipped or copper-nickel-chromium finished, anodized aluminum, or baked enamel on electro zinc plated or phosphatized steel.

3.4.14 Platters. Unless otherwise specified (see 6.2), meat platters, approximately 10 inches by 30 inches (254 mm by 762 mm), in sufficient quantity to cover the shelf and floor rack of the storage area in type III double-duty cases shall be furnished with each type III double-duty display case. Unless otherwise specified (see 6.2), the platters shall be anodized aluminum or stainless steel.

* 3.4.15 Shelves for display case. The shelving located in the upper part of class 2, 3, 4, 5 and 6 display cases shall be baked enamel on steel or aluminum, stainless steel or aluminum. Color of applied finishes shall be white. Shelves and supports shall be designed to support the loading specified for superstructure shelves (see 3.5.6 and 4.4.4). All shelves shall have a full length tag moulding to take 1-1/4-inch (32 mm) tags located on the front of the shelves.

3.4.16 Deck racks. Deck racks for display cases shall be baked enamel on electro-zinc plated or phosphatized steel, stainless steel, anodized aluminum alloy, aluminum, hot tin dipped steel, or copper-nickel-chromium finished steel.

3.4.17 Evaporators. The evaporators shall be constructed of copper or aluminum tube with either copper or aluminum fins. Fins shall be bonded to tube to assure a tight bond of high conductivity. Evaporator shall be dehydrated and sealed. Refrigerant connections shall be easily accessible for servicing.

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* 3.4.17.1 Fans. Fans shall provide positive forced circulation of the refrigerated air to maintain uniform temperature throughout entire display area to be refrigerated. The fan and hub shall be made of electro-tin plated brass, aluminum alloy, or stainless steel. The fan shall be connected direct to the motor. The motor and fan shall be accessible for servicing or replacement. When specified (see 6.2), a fan kit shall be provided for use at the base for back to back placements of display cases.

3.4.18 Condensing units. The display cases shall have air cooled, remote or self-contained condensing units, or water cooled, remote or self-contained condensing units, as specified (see 6.2). The condensing unit shall consist of a motor, compressor, condenser, receiver, motor safety devices, and other necessary equipment all mounted and arranged on a unit base or rack. The unit base or rack shall be shock mounted to reduce vibration to a minimum. Unless otherwise specified (see 6.2), condensing units for self-contained cases designed for 50- or 60-hertz operation, and remote units designed for 60-hertz operation, shall be hermetic, either thermally or mechanically sealed. Remote units designed for 50-hertz operation may be open type or hermetic, thermally or mechanically sealed. The condensing units procured for operations on a frequency of 50-hertz shall operate with full load for 4 hours at the specified frequency and at 90 percent of the specified system voltage followed by operation for 4 hours at specified frequency and at 110 percent of specified voltage, and shall not cycle on the motor protector (see 4.4.1.2c). All condensing units shall be piped, valved, wired, and serviced ready for operation and connection to utility sources as required. The assembled unit shall be dehydrated and sealed; remote units shall contain a holding charge, and self-contained units a full operating charge of the specified refrigerant. The compressor shall contain the proper amount of manufacturer's recommended oil. When multiple display case lineups are to be operated by one condensing unit, not over 48 feet (15 m) of display case shall be operated by one condensing unit. The refrigerant shall not contain moisture in excess of 10 parts per million (P.P.M.) (see 4.4.4).

3.4.18.1 Condensing unit base and enclosure. Condensing units for self-contained cases shall be mounted on a unit base and for remote cases shall be mounted on a unit base or rack. The unit base and rack shall be constructed of structural steel shapes welded to form a rigid structure. The condensing units, control panel, and accessories complete and ready for connection as specified in 3.4.18 shall be mounted on the unit base or rack. When specified (see 6.2), the condensing unit rack designed for remote installation shall be furnished with a metal enclosure to house the condensing units to permit locating the condensing units outside of the building. The housing to enclose the condensing unit shall be constructed as a removable enclosure with separate steel frame, and hinged panels so that condensing units may be easily exposed for operation, maintenance, and servicing. Enclosure shall be vented.

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3.4.19 Controls and accessories. All controls and accessories required for satisfactory operation of the display case, when connected to the condensing unit, shall be supplied and shall be piped and wired and ready for use. The refrigeration system shall include adjustable thermostatic expansion valves, heat exchanger as required, dehydrator, dual pressure control with adjustable differential, compressor service valves, oil separator if necessary, liquid line receiver shut-off valve, combination moisture and liquid indicator, temperature alarm system, and control devices such as demand defrost controls and dewpoint controllers as necessary for effective automatic operation and defrosting. A time-pressure or time-temperature clock electric defrosting system for time initiated and pressure or temperature terminated defrost shall be provided. Refrigeration shall be restarted automatically by pressure or temperature. The clock shall include a safety device which shall automatically terminate the defrost period by time if the pressure or temperature fails to reach the desired cut-in point. A thermometer shall be mounted to show the temperature of the air leaving the display area and positioned to be clearly visible from the center front of the display case. Case temperature shall be controlled by pressure settings at the condensing unit. Any transformers required for operation of the control circuits or lighting and cooling fan motors on 120-volt electrical current shall be furnished with the units supplied.

3.4.20 Motors. Unless otherwise specified (see 6.2), motors rated under 3/4 horsepower shall be designed for operation on nominal 120-volt, 1-phase, 60 hertz alternating current electrical system, and motors rated 3/4 horsepower or over shall be designed for operation on nominal 240-volt, 3-phase, 60-hertz alternating electrical system. Compressor motors shall have sufficient capacity to start the refrigerating system under full load and to operate in high ambients specified without cutting out on overload. When frequency other than 60-hertz is specified, the motor shall be specifically designed for the frequency cited and the frequency shall be included with the data on the nameplate.

3.4.20.1 Motor starter. Automatic starters shall be supplied for aall 3-phase motors and for all 1-phase motors over 1 horsepower (7 W). The compressor motor starter shall be across-the-line type, continuous duty, suitable for automatically starting the condensing unit compressor motor when actuated by the automatic controls. The starter shall be mounted in a drip-proof enclosure. The motor starter shall be equipped with a thermal overload protected if the motor is not otherwise equipped with such protection. All relays shall be the "quick-trip" type, factory calibrated for the particular motor application.

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3.5 Finish.

* 3.5.1 Display case exterior. The exterior of the display case shall be finished in baked synthetic enamel, as standard with the manufacturer and shown in current published catalogs and sales literature. Unless otherwise specified (see 6.2), the exposed top, fronts, back and ends shall be white. The bottom of cabinets, and the backs of units specified for wall installation shall be aluminum or one of the materials specified in 3.4.8.1 unpainted or a black or white color. The kick-plate or splash guard shall be black or in a color and finish to match the trim. Trim shall be manufacturer's standard in black baked porcelain enamel, stainless steel, or aluminum alloy finish standard with the manufacturer.

3.5.2 Display case interior. The interior of the display case shall be stainless steel or aluminum alloy finish standard with the manufacturer; baked enamel on phosphatized steel; or any combination of the foregoing unless otherwise qualified by these specifications. The interior front and bottom may be unpainted minimum spangled galvanized steel. Manufacturer's recommended load line shall be clearly marked and identified in a contrasting color. In double-duty cases, the interior liner of the storage area shall be the finish standard with the manufacturer.

3.5.3 Superstructures. All components of the superstructures including ends, backs, tops, pilasters, structural supports, shelves, braces, dividers, and accessories shall be finished in baked enamel and fitted with trim to match the display case.

3.5.4 Application.

* 3.5.4.1 Baked enamel. Baked enamel finishes shall be a synthetic baked enamel applied to surfaces that have been cleaned and primed. The finish shall be smooth, applied uniformly, durable, tightly adhering, and non-absorbent. There shall be no visible corrosion when the finished surfaces are wiped with a damp cloth after being exposed to a salt spray for 96 hours (see 4.4.5).

* 3.6 Marking for identification. Display cases and condensing units shall be marked with the following information on nonferrous metal plates. Marking shall be applied to the plate by metal stamp, vibro peening, acid, electric or electro-chemical etching, or embossing.

1. Nomenclature
2. Type, style, class, ambient temperature operation, superstructure and size (see 1.2).
3. National stock number.
4. Manufacturer's model, type, or catalog number.
5. Serial number.
6. Contract or order number.
7. Other information required by UL Standard No. 471.

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3.7 Wiring diagram. A wiring diagram shall be included with each instruction manual.

* 3.8 Workmanship. All metal forming shall be done with dies or fixtures to insure uniformity of size and shape. All sharp corners, edges, or projections on the interior or exterior shall be eliminated. Sheet metal or structural sections used in fabrication of the units shall be free of kinks, bends, and winds. Shearing and chipping shall be done accurately and cleanly on all portions of the work and shall be neatly finished. All bolt and rivet holes shall be accurately and cleanly punched or drilled. Rivets, when used, shall be driven to completely fill the holes, with full-sized heads neatly finished and in full contact with the surface of the member. Bolts, nuts, and screws shall be drawn tight and so secured that they will not work loose under vibration. Optional equipment, accessories, and piping shall be securely fastened and braced to prevent damage or loosening during transit or operation.

3.9 Servicing after testing. Upon completion of testing, the display case shall be allowed to come to equilibrium at room temperature. The interior and exterior shall be dried, cleaned, inspected, and required adjustments, including setting temperature controls for specified operating conditions, shall be made prior to final preparation for packing and packaging. Condensing units shall be disconnected, piping sealed, and the condensing units prepared as specified in 3.4.18.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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* 4.2.2 In-process inspection.

* 4.2.2.1 Visual examination of operations. Examination shall be made at any point or during any phase of the manufacturing process to determine that the required operations or assemblies are accomplished as specified. Whenever nonconformance is noted, corrections shall be made on affected items and process.

(a) Determine reinforcement and backing of shell and liners is provided as well as predrilled, (see 3.4.7).
(b) Determine that fitted covers of the access openings are in place (see 3.4.8).
(c) Determine that insulation is installed as specified in 3.4.8.
(d) Attachment of backing plates, tapping plates, and reinforcing to determine compliance with 3.4.8.2.
(e) Application of baked enamel finish (see 3.5.4.1).
(f) Punching and drilling of bolt and rivet holes (see 3.8).
(g) Piping is fastened or braced (see 3.8).
(h) All metal forming is as specified (see 3.8).
(i) Shearing and chipping accurate and clean (see 3.8).

4.2.2.2 In-process tests. Any in-process testing failure shall be cause for rejection of all components represented by the applicable sample.

* 4.2.2.2.1 Organic enamel test. The salt spray test as specified in 4.4.5 shall be performed (on test specimens prepared at the time) the initial production unit is fabricated. The size of the sample specimens shall be 3 inches by 6 inches (76 mm by 152 mm). As a minimum, one representative sample of each component being finished with baked enamel shall be tested.

4.2.2.2.2 Testing of density of insulation. The density test for insulation shall be performed as specified in 4.4.6. The inspection lot shall be the amount of material used in one day's production. The sample size for this test shall be a 6-inch (150 mm) cube of insulation.

4.2.3 End item inspection. The sample unit for this examination shall be one completely fabricated display case. The inspection lot shall consist of all display cases of one type, style, class, size, ambient temperature operation, and superstructure offered for inspection at one time.

4.2.3.1 Visual examination. Examination of the display cases shall be in accordance with the classification of defects in table IV. The inspection level shall be level II, with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in defects per hundred units.

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TABLE IV. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not color specified	X	
	Not finish specified	X	
	Load line not as specified	X	
Design, construction and workmanship (applicable to all components and assemblies)	Part missing, not in proper alignment, or not type specified	X	
	Adjustable assembly that cannot be adjusted or is not properly adjusted to perform function intended (condition cannot be readily rectified)	X	
	Adjustable assembly not properly adjusted but can be adjusted		X
	Component not attached as specified	X	
	Component not accessible for servicing where required		X
	Access openings not as specified	X	
	Sharp edges, burrs, or slivers not removed	X	
	Bolts, nuts, screws, studs, or other type of threaded fasteners, missing, broken, stripped, fractured or loose		X
	Sheet metal has kinks, bends or winds		X
Rivets (where applicable)	Rivets not completely driven, heads not full size, not in contact with both surfaces		X
Wiring and electrical	Wiring not as specified	X	
	Wiring diagram missing, incomplete, not legible, not affixed in prescribed manner		X
	Lighting not as specified	X	
Marking for identification	Missing, incomplete, not legible, not specified type or size, not in proper location, plate not as specified		X

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4.2.3.2 Dimensional examination. Examination shall be made of the end item to determine compliance with dimensional requirements. Any dimension that does not meet specified requirements shall be classified a defect. The inspection level shall be S-2 and the AQL shall be 6.5, expressed in terms of defects per hundred units.

* 4.2.3.3 Testing of the end product. Each production unit shall be tested as specified in 4.4.4. Any nonconformance shall be cause for rejection of the unit. One display case from the lot shall be tested as specified in 4.4.1 thru 4.4.3. Any nonconformance shall be cause for rejection.

* 4.2.4 Inspection of packaging. An inspection shall be made to determine the preservation-packaging, packing, and marking are as required by section 5. Defects shall be as indicated in table V. The sample unit shall be one container fully prepared for inspection. The lot shall be the number of containers offered for delivery at one time. The inspection level shall be S-2 with AQL of 6.5, expressed in terms of defects per hundred units.

TABLE V. Packaging inspection

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application
Preservation	Preservative improperly applied or missing Floating mechanism not secured All components and packages not positioned or secured as specified
Materials	Component missing, damaged or otherwise defective

4.2.4.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-621 or MIL-C-104, examination for defects in anchoring, closure, and strapping shall be in accordance with the applicable container specification.

4.2.4.2 Examination of interior packages. When packages are required to comply with PPP-B-636, examination for defects in the closure shall be in accordance with the appendix of PPP-B-636.

4.3 Certification. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

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* 4.4 Test method. Prior to approval of the first shipment, the contractors may submit, in lieu of the testing requirements specified in 4.4.1 thru 4.4.7, certified test data and test results for each type, style, and size display case to be delivered under the contract. The test results shall show, to the satisfaction of the contracting officer, that the display cases will conform to the test requirements of this specification. In addition, if the contract requires delivery of a display case assembly consisting of two or more of the same type and style display cases, and the assembly is to be operated by one or more condensing units, the contractor shall certify that the capacity of the condensing unit(s) to be furnished with each assembly (line-up) shall equal or exceed the combined refrigeration capacity requirements of all display cases in the assembly (line-up).

* 4.4.1 Performance test.

4.4.1.1 Display case preparation.

(a) The display case with the refrigeration system shall be assembled and set up for the performance test of 4.4.1.2. All shelves, racks, plugs, and partitions shall be in place in accordance with the requirements specified.

(b) The refrigeration system shall be charged; the refrigeration unit shall be given a run-in operating period of approximately 8 hours at room temperature; controls shall be checked; and the power supply voltage adjusted to ± 5 percent of the voltage specified for design and operation, and other electrical characteristics shall be as specified.

(c) Placement of the display case in the test room shall be made to simulate conditions in an installed location, that is, wall cases shall be backed up to a wall space, the ends closed off with a temporary partition, and at least 5 feet (1.5 m) of clear space provided at the front of the case. Center aisle cases shall have at least 5 feet (1.5 m) of clear space all round the display case.

(d) Air circulation in the room shall be such that the specified uniformity of temperature in the heated room is obtained. The display case shall be shielded from forced air currents. Ambient temperatures shall be measured at points 1 foot (0.3 m) above the floor and 3 feet (0.9 m) above the cabinet at the center point on each side. Ambient temperatures at each point shall be maintained within ± 2°F (± 1°C) of the specified test conditions. The test conditions shall be maintained during stabilization periods as well as during test runs. Where cases are adjacent to walls or partitions, the temperature of such walls and partitions shall be the same as the room temperature.

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(e) In preparation for any operating test, the display case shall be operated under specified test conditions for sufficient time to establish steady or repetitive state. Immediately after this preparatory period, the display case shall be operated for the specified test period during which the specified elements of performance are determined and recorded. Temperature observations shall be made continuously or at equal intervals of not greater than 15 minutes. Reported temperatures shall be the arithmetical average of all readings taken at equal time intervals.

(f) The display case (display area and storage area) shall be loaded with simulated load corresponding to the manufacturer's recommended maximum load.

(g) The interior display case temperature shall be measured with not less than eight thermocouples or other temperature measuring devices, located in the top of the interior load and equally distributed across the top of the display compartment. Double-duty cases shall have four additional thermocouples equally distributed throughout the load in the bottom storage compartment.

(h) The condensing unit of remote cases shall be located at least 5 feet (1.5 m) from the display case with the refrigerant lines uninsulated and specified ambient air entering the condenser. Condenser inlet air temperature of both remote and self-contained condensing units shall be determined by averaging the readings of four temperature measuring devices located 1 to 2 inches (25 to 51 mm) from the condenser fins and equally distributed over the inlet air area.

(i) The electric defrosting device shall be made inoperative for the performance test.

(j) Prior to the start of the performance test, the display cases may be precooled to the temperatures specified below:

Type	Precooled temperature
I	32°F (0°C)
II	31°F (-0.5°C)
III	28°F (-2°C)
IV	-10°F to -20°F (-23°C to -29°C)
V	-18°F to -25°F (-28°C to -32°C)

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(k) Where thermometers are used to obtain temperature data for performance requirements, they shall be numbered and each one shall be located in a fixed position for the duration of the test, and shall not be moved except as required to take readings at the fixed position. Thermometer readings for either bulb or glass type, shall be made to an accuracy of 1/2 a thermometer graduation.

4.4.1.2 Procedure.

(a) The condensing unit of the display case shall be placed in operation when the condensing unit ambient temperature is 100°F (38°C) for remote units. The display case ambient temperature, determined as the average of the readings of the temperature measuring devices, is 78°F (25°C) and 50 percent relative humidity or 90°F (32°C), as specified. During the test period, observations shall be made that the interior display case temperature is as specified in table III, determined as the average of the temperature measuring devices located in the display case. An observation shall be made that no single interior temperature (regardless of average) and none of the running times shall exceed the interior temperature and running time requirements specified in 3.4.4.2. Types IV and V self-contained cases may be tested with night covers specified in 3.4.4.2 in place for the 90°F (32°C) ambient requirement.

(b) Upon satisfactory completion of the controlled temperature display case performance tests, the remote condensing unit ambient temperature shall be raised to 110°F (43°C) while the display case ambient temperature is held constant. Ambient temperatures for self-contained display case tests shall be raised to 100°F (38°C). The display cases shall be observed for a period of 4 hours at these elevated temperatures without cutting out on overload to conform to the overload requirements of 3.4.4.2. During operation in the elevated ambient temperatures, the contractor may return the electric defrost devices to operating condition, and the internal display case increased temperatures shall be taken every 30 minutes and made a matter of record in the test data compilation.

(c) Refrigeration units procured for operations on a frequency of 50 hertz shall be tested for conformance with 3.4.18. Ambient display case conditions for this test shall be normal room temperature, 60°F to 100°F (15°C to 38°C). The motor shall not be damaged and shall not cut out on either inherent or external overload protectors during these tests.

4.4.2 Exterior temperature test. To determine exterior surface temperature conformance to 3.4.4.2, the display case shall be operated for a period of not less than 4 hours under the conditions specified in 4.4.1.2. During and at the end of the 4 hour test the entire exterior of the display case shall be inspected in 14-foot (4.3 m) quadrants to determine that exterior surfaces are not more than 10°F (6°C) below room ambient test temperatures.

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4.4.3 Shelf load tests. Canopies and shelves for the display case and superstructures shall be assembled and mounted as intended for use in their service installation. To determine uniform load compliance with 3.4.6 and 3.4.15, a load of 50 pounds per square foot (244 kg/m^2) shall be distributed uniformly on the entire top area of all canopies and on all shelf areas. Material for loading shall be nonrigid and shall be in segments with a base area of not more than 72 square inches (0.046 m^2). When fully loaded it shall be observed that the canopies and shelves shall show not more than 1/4-inch (6 mm) deflection from the horizontal, and upon removal of load, the supports, canopies and shelves shall show no deformation or permanent set.

4.4.4 Operation test. Display cases and condensing units shall be operationally tested, electrically tested, and leak tested, to determine compliance with operation requirements of 3.4.4.3. Display cases for remote condensing units need not be connected to a condensing unit nor perform any refrigeration function during this test. During and after this test, for self contained cases, the combination moisture and liquid indicator shall be observed and shall indicate "dry" within the limits specified in 3.4.18. A liquid line temperature of not more than 100°F (38°C) shall be maintained when observing the indicator. If the indicator shows that there is moisture in the system, drier replacements shall be made and the system operated until the moisture indicator shows dry.

* 4.4.5 Salt spray test. Metal test specimens, coated with organic material shall be subjected to salt spray test in accordance with ASTM B 117 for 96 hours. The test specimens shall be representative of components of the display case. After exposure to the salt spray for 96 hours, the surface of the specimens shall be wiped with a damp rag and an examination for corrosion made to determine compliance with 3.5.4.1.

4.4.6 Density test for insulation. When rigid or pre-expanded insulation is used a 6-inch (150 mm) cube shall be cut and weighed. This weight, 1/8 of a cubic foot (1/64 of a cubic meter), shall be multiplied by 8 (64) to determine conformance to the density requirements of 3.3.1.1. When foamed-in-place insulation is used, one free foam sample of the insulation specified in 3.3.1.2, a minimum size 8-inch (0.3 m) cube, shall be taken prior to each day's production. From the center of the sample, a 6-inch (150 mm) cube shall be cut and weighed. This weight, 1/8 of a cubic foot (1/64 of a cubic meter), shall be multiplied by 8 (64) to determine conformance with the density requirement of 3.3.1.2. In addition a daily record of the foam insulation shall be kept as follows and the production unit shall not deviate more than ± 5 percent from the data recorded on the unit used for the test specified in 4.4.2.

Calibration of mix (mix proportions)
Amount of flow (per unit)
Time of flow (per unit)
Mould temperature (per unit)

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservative. The refrigeration system shall be prepared for shipment in accordance with the manufacturer's recommended procedure for overseas shipment or as otherwise specified. Drive belts shall be removed, and exposed uncoated ferrous metal surfaces of pulleys and shafts shall be cleaned in accordance with process C-1 of MIL-P-116 and coated with primer conforming to TT-P-664. When the primer is thoroughly dry, drive belts shall be remounted in place with tension on the belts relaxed. All floating mechanisms and tubing not rigidly fixed, such as vibration loops, free coils, capillary tube loops and coils shall be secured in a fixed position to minimize vibration and subsequent damage during handling and shipment. Floating mechanism is defined as any part that is inherently flexible or movable, or any part that is rubber, spring, bracket, or otherwise mounted which might vibrate during shipment.

5.1.1.2 Unit packing.

5.1.1.2.1 Removable components.

5.1.1.2.1.1 Mirrors, glass, and glazed doors. When removable mirrors, glass, and glazed doors are supplied, they shall be removed and individually unit packed in corrugated fiberboard boxes conforming to style FOL, type CF, variety SW, class domestic of PPP-B-636. The opening shall be on the panel with the longest dimension and the adjacent shortest dimension. A minimum of 1-1/2 inch (38 mm) clearance shall be maintained between the sides of the glass or mirror and the interior surfaces of the fiberboard box by means of spring pads made of the same material as the box, placed at each edge of the mirror, glass, or glazed door. Box closure shall be in accordance with the appendix of PPP-B-636. The boxes containing the mirrors, glass, or glazed doors shall be positioned on edge (vertical position) and shall be blocked or braced within the well of the display case or inside the shipping container in a manner to prevent shifting during shipment and handling.

5.1.1.2.1.2 Shelves, racks, platters, or trays. The removable shelves, racks, platters, or trays shall be unit packed in a corrugated fiberboard box or boxes conforming to style RSC or OPF, type CF, variety SW, class domestic of PPP-B-636. Box closure shall be in accordance with the appendix of PPP-B-636. The fiberboard box or boxes shall be placed within the well of the display case and shall be immobilized in a manner to prevent damage or shifting during shipment and handling.

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5.1.1.2.1.3 Superstructures. Superstructures which are normally demounted for shipment shall be placed in the well of the display case. The end mounting bolts of the display case shall be fitted with spacers to permit free entry of the superstructure within the interior of the case. The superstructure shall be immobilized in the well of the display case in a manner to prevent shifting during shipment and handling. All blocking and bracing in direct contact with the display case or superstructure shall be cushioned with cellulose wadding or fiberboard to prevent abrasion of enameled or porcelained surfaces.

5.1.1.2.2 Fixed components and components not normally removed for shipment.

5.1.1.2.2.1 Fluorescent bulbs. All installed fluorescent bulbs shall be secured in place with pressure-sensitive tape.

5.1.1.2.2.2 Glass, mirrors, and glazed doors. All fixed or nonremovable glass, mirrors, and glazed doors shall be protected in a manner to prevent breakage during shipment and handling.

5.1.1.2.2.3 Shelves and racks. Shelves and racks normally shipped in place shall be secured in place in a manner to prevent damage in shipment and handling.

* 5.1.1.2.3 Miscellaneous parts, instruction manual, and assembly hardware. All miscellaneous component parts and assembly hardware shall be unit packed method III of MIL-P-116. The packaged components and assembly hardware shall be secured in place within the well of the display case in a manner to prevent damage or shifting during the shipment and handling.

5.1.2 Commercial. Each display case shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A or Commercial, as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Display case. Each display case, preserved as specified in 5.1, shall be packed in a sheathed, nailed wood crate conforming to type I, class 1 or 2, style a or b of MIL-C-104. The boxes containing doors, not positioned within the well of the display case shall be positioned on edge (vertical position) and secured within an unused portion of the shipping container. Anchoring and cushioning of contents shall be in accordance with the appendix of MIL-C-104.

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5.2.1.2 Condensing units. Each remote condensing unit shall be packed in a nailed wood shipping container conforming to class 2, style 2 or 4, grade A or B, type 3 load of PPP-B-621. Each shipping container shall be provided with skids in accordance with PPP-B-621. Units shall be secured within the shipping container with standard carriage or step bolts. The size of the bolt shall be determined by the diameter of the hole in the base of the item. A lockwasher shall be used under the nut. Bolts securing the unit in the shipping container shall pass through the skids; bolting through the base only is prohibited. Bolt heads shall be countersunk into the skid runners and no portion of the bolt head shall extend below the bottom surface of the skid runner. A minimum of one inch (25 mm) clearance shall be provided between the unit and the inside surfaces of the side, end, and top panels of the shipping container. Each shipping container shall be provided with a type I or II, grade C case liner conforming to MIL-L-10547 or a full height shroud made from material conforming to PPP-B-1055. All seams of the shroud shall be sealed with a 3/4-inch (19 mm) minimum width continuous seam of water-resistant adhesive conforming to MMM-A-260. Each shipping container shall be closed and strapped in accordance with the appendix of PPP-B-621.

* 5.2.2 Commercial packing. Each display case, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

6. NOTES

6.1 Intended use. The display cases covered by this specification are intended for use in establishments dispensing food, for the refrigerated display of products.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Type, style, class, size, suffix designation, superstructure, and ambient temperature operation of display case (see 1.2)
(c) When glass other than tempered glass or safety glass is required (see 3.4.1)
(d) Information required for matching lineup (see 3.4.3)
(e) When 78°F (25°C) or 90°F (32°C) ambient operating temperature is required (see 3.4.4.2)
(f) When access doors to storage space shall be in the front or rear of display cases (see 3.4.5.1)
(g) Type and size of ends required for style F display case (see 3.4.5.7)

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(h) When adjustable wire shelves are required (see 3.4.5.7)
(i) When a joint kit is required to join two or more style F display cases (see 3.4.5.7)
(j) When a removable shelf kit is required (in lieu of roll-in racks) (see 3.4.5.8)
(k) Size of platters if other than specified (see 3.4.14)
(l) Whether material for platters is other than specified (see 3.4.14)
(m) When a fan kit is required (see 3.4.17.1)
(n) Whether condensing units shall be air cooled, water cooled, remote or self contained (see 3.4.18)
(o) When other than hermetic type condensing units are required (see 3.4.18)
(p) Whether condensing unit enclosure is required (see 3.4.18.1)
(q) Electrical characteristics if other than specified (see 3.4.20)
(r) Color, if other than white (see 3.5.1)
(s) Selection of applicable levels of preservation and packaging, and packing (see 5.1 and 5.2)

6.3 Definitions. The following definitions are applicable to this specification:

Accessible: Accessible shall mean readily exposed for proper and thorough cleaning and inspection with the use of only simple tools, such as a screwdriver, pliers, or open-end wrench.

Cleaning: The term cleaning shall mean the physical removal of residue of dirt, dust, foreign material, or other soiling ingredients or materials.

Readily cleanable: Readily (or easily) cleanable shall mean readily accessible and of such material, finish, and fabrication that soil may be effectively removed by normal cleaning methods.

Readily removable: Readily (or easily) removable shall mean capable of being taken away from the main unit without the use of tools.

Smooth: The word "smooth" is used to define a surface free of pits and inclusions and a cleanability equal to commercial grade hot rolled steel free of visible scale.

Sealed: Spaces required to be "sealed" shall have no openings that shall permit the entry of insects, rodents, dirt, or moisture seepage.

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6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

* 6.5 Changes from previous issue. The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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